MILITARY SPECIFICATION SHEET

FILTER ELEMENT, FLUID PRESSURE, HYDRAULIC, AIRCRAFT, DISPOSABLE 5 MICRON ABSOLUTE

This specification sheet is approved for use by the Department of the Air Force and is available for use by all departments and agencies of the Department of Defense.

The complete requirements for acquiring the filter element described herein shall consist of this specification and the latest issue of specification MIL-F-83861.

REQUIREMENTS

Bubble point. The initial and burst out bubble point values shall meet the values established by the activity responsible for qualification and shall be determined as specified in 4.7.1.1 of general specification.

Filtration. The filter elements shall remove 100 percent of all contaminants over 5 microns in size and 95 percent by weight of add contaminants.

Cold start. The filter elements shall withstand 3000 psi differential PRESSURE across the element for the flow cycles and fluid temperature as specified in 4.7.3 of general specification.

Element pressure drop. The pressure drop through the clean element at rated flow shall not exceed the values specified. Test fluid shall be maintained at 100 ± 10°F.

Part number	Rated flow	Pressure drop
M83861/3-1	6.0 gpm	15.0 psid
M83861/3-2	16.0	15.0
M83861/3-3	29.0	15.0
M83861/3-4	29.0	10.0

Element dirt capacity. The filter element shall meet the dirt capacities specified in Table I, Contaminants.

Element collapse. The filter element shall withstand differential pressures specified:

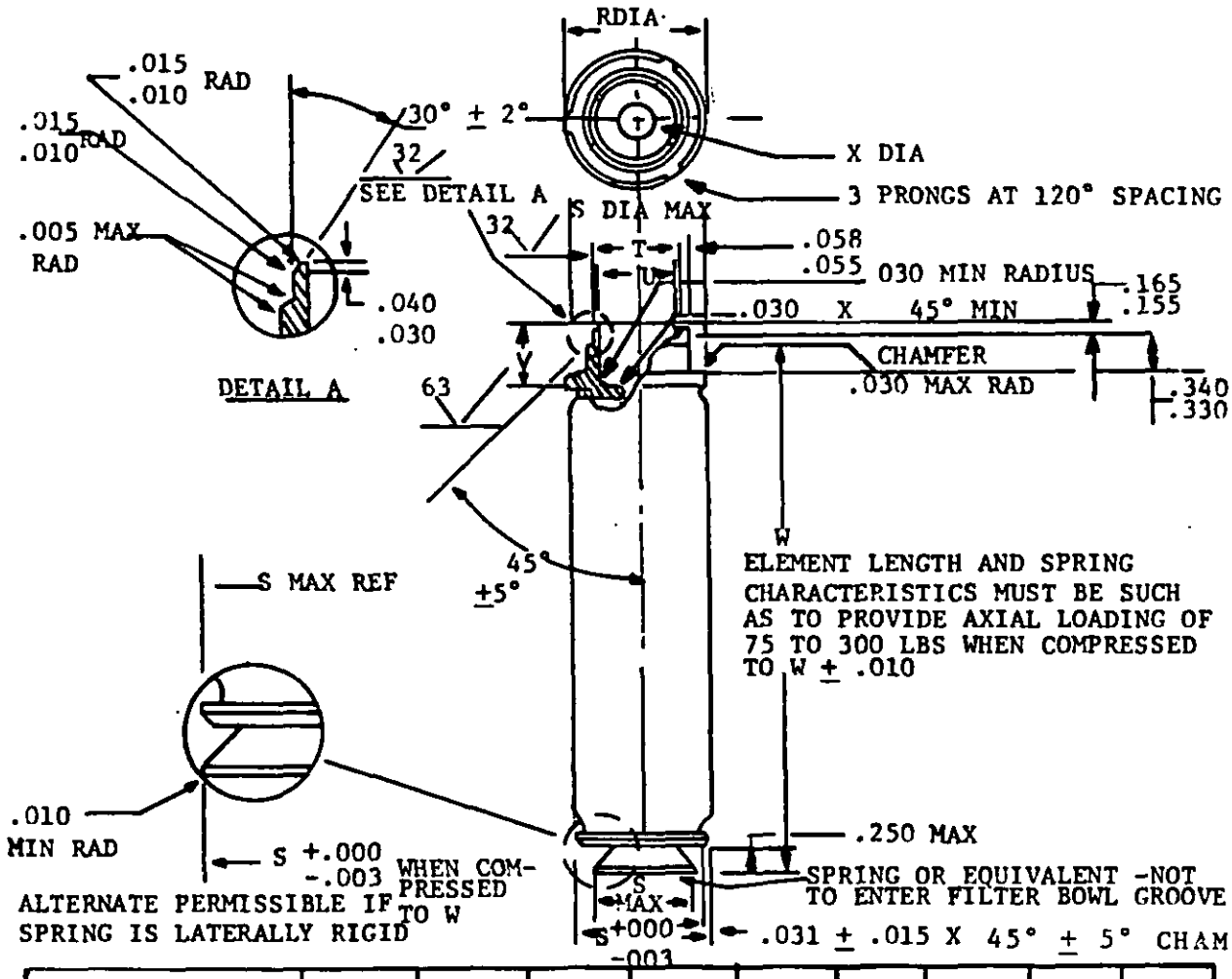
M83861/3-1	4500 psid
M83861/3-2	4500 psid
M83861/3-3	4500 psid
M83861/3-4	300 psid

Description. The filter element shall be as shown in Figure 1 as specified herein.

ASMC N/A FSC 1650

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Part Number	RATED FLOW GPM	R +.004 -.010	S	T ±.001	U MIN	V MIN	W	X MIN	MAX WT LBS
M83861/3-1	6.0	1.250	1.137	.748	.670	.490	2.980	.435	0.35
M83861/3-2	16.0	1.625	1.500	1.122	1.035	.690	4.220	.715	0.83
M83861/3-3	29.0	1.985	1.845	1.372	1.155	.760	5.520	.995	1.50
M83861/3-4	29.0	1.985	1.845	1.372	1.155	.760	9.047	.995	1.50

FINISH 100 ALL OVER EXCEPT AS NOTED
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES DECIMALS ± .010, ANGLES ± 1°.
CONCENTRICITY: SURFACES OF DIAMETERS T, R AND S MUST BE CONCENTRIC WITHIN .012 TOTAL INDICATOR READING.

FIGURE 1. Filter element, 5 micron absolute.

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Flow fatigue. The filter elements shall withstand the flow fatigue cycles specified at 300 cpm.

P/N	Cycles	Pressure drop
M83861/3-1,-2,-3	1,125,000	20
	112,000	50
	12,500	100
	400	1500
	100	3000
M83861/3-4	1,125,000	20
	112,000	50
	12,000	100
	500	300

Media migration. Element migration when tested in accordance with 4.7.8.3.2 of general specification shall not exceed 0.0040 grams and there shall be no migration identifiable as coming from the filter media.

Maximum particle passed. The maximum particle passed shall be no greater than 5 microns.

TABLE I. Contaminants.

		Degree of filtratio test		Maximum particle test		Pressure buildup test
Dash No.	Flow rate (gms)	Add rate F-9 beads (gms)	Add rate AC dust (gms)	Add rate Carbony 1 (gms)	Add rate AC dust (gms)	Minimum AC dust (gms)
-1	6	0.15	0.15	0.30	0.2	0.8(90psid)
-2	16	0.15	0.15	0.30	0.4	2.4
-3	29	0.15	0.15	0.30	1.0	5.0
-4	29	0.15	0.15	0.30	1.0	15.0

Part number. The part number consists of letter 'M' plus basic number of this specification sheet and a dash number.

Example: M83861/3-1

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 71

Project Number
1650-F132

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